Usefulness of Muscle Synergy Analysis in Individuals With Knee Osteoarthritis During Gait

K. Kubota, H. Hanawa, M. Yokoyama, S. Kita, K. Hirata, T. Fujino, T. Kokubun, T. Ishibashi, and N. Kanemura

***Abstract*— *Objective:* To clarify whether there are any muscle synergy changes in individuals with knee osteoarthritis, and to determine whether muscle synergy analysis could be applied to other musculoskeletal diseases. *Methods:* Subjects in this study included 11 young controls (YC), 10 elderly controls (EC), and 10 knee osteoarthritis patients (KOA). Gait was assessed on a split-belt treadmill at 3 km/h. A non-negative matrix factorization (NNMF) was applied to the electromyogram data matrix to extract muscle synergies. To assess the similarity of each module, we performed the NNMF analysis assuming four modules for all of the participants. Further, we calculated joint angles to compare the kinematic data between the module groups. *Results:* The number of muscle modules was significantly lower in the EC (2–3) and KOA (2–3) groups than in the YC group (3–4), which reflects the merging of late swing and early stance modules. The EC and KOA groups also showed greater knee flexion angles in the early stance phase. Contrarily, by focusing on the module structure, we found that the merging of early and late stance modules is characteristic in KOA. *Conclusion:* The lower number of modules in the EC and KOA groups was due to the muscle co-contraction with increased knee flexion angle. Contrarily, the merging of early and late stance modules are modular structures specific to KOA and may be biomarkers for detecting KOA. *Significance:* Describing the changes in multiple muscle control associated with musculoskeletal degeneration can serve as a fundamental biomarker in joint disease.**

***Index Terms*— Electromyography, knee osteoarthritis, muscle co-contraction.**

I. Introduction

ABNORMALITIES of the neuromuscular system including muscle strength loss occur with aging [1], [2]. The progression of these abnormalities leads to pathological conditions such as osteoarthritis [3]. In particular, the knee joint is one of the most complex structures in the human body that undergoes large motions. Therefore, deviation from normal joint motion contributes to the development and progression of knee osteoarthritis (KOA). To prevent this abnormality, it is important to comprehensively verify the difference in the neuromuscular system between healthy and elderly individuals and those with KOA. KOA can be defined through symptoms or pathological findings, especially pathological assessments characterized by radiographic features such as joint space narrowing, subchondral sclerosis, and marginal osteophytes [4]. These altered structures and mechanics of the knee joint induce joint instability and dysfunction. To stabilize the knee joint during gait, KOA involves greater muscle co-contraction. While these alterations are believed to be attempts to reduce pain and protect the knee from further degeneration, in the long term, they may have adverse effects on the knee joint [5].

Additionally, several studies reported the relationship between aging and increased muscle co-contraction [6]–[8]. Muscle weakness associated with aging required multiple muscle activity to exert the proper torque [9]. However, age does not necessarily lead to KOA. Therefore, it is essential to understand the changes in the neuromuscular system specific to KOA based on the muscle activity patterns between older adults and those with KOA.

Recently, the muscle synergy analysis [10] has been used as a method to clarify the alteration of the neuromuscular system. The muscle synergy describes a potential mechanism by which the nervous system can simplify the control of a large number of muscles [11], [12]. In this method, multi-channel electromyography was used to evaluate the quantitative activity patterns of multiple muscles. Many researchers have investigated the muscle synergy associated with gait, and three to five synergies were defined from over 16 muscles [13], [14]. The more primitive elements of the motor system, such as muscle synergy, have robustness, and these are hard-wired into the motoneuronal network involved in gait control [15].

Since the spinal cord [16] and cerebral cortex [17] are involved in the recruitment of muscle synergy, many previous studies have focused on neurological disorders [18]–[21]. Contrarily, muscle co-ordination patterns and muscle synergy are constrained by the biomechanics of the musculoskeletal

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K. Kubota, M. Yokoyama, and S. Kita are with the Graduate Course of Health and Social Services, Graduate School of Saitama Prefectural University, Koshigaya 343-8540, Japan.

H. Hanawa and T. Fujino are with the Department of Rehabilitation, Faculty of Health Sciences, University of Human Arts and Science, Saitama 339-8539, Japan.

K. Hirata is with the Department of Rehabilitation, Faculty of Health Sciences, Tokyo Kasei University, Sayama 350-1398, Japan.

T. Kokubun and N. Kanemura are with the School of Health and Social Services, Saitama Prefectural University, Koshigaya 343-8540, Japan (e-mail: kanemura-naohiko@spu.ac.jp).

T. Ishibashi is with the Ageo Futatumiya Clinic, Ageo 362-8588, Japan.

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TABLE I
THE PARTICIPANT DEMOGRAPHICS

		Mean	SD	Range
YC group (n = 11)				
Age (yr)		20.5	1.8	20-26
Sex (male/female)	11/0			
EC group (n = 10)				
Age (yr)		70.0	5.0	61-77
Sex (male/female)	8/2			
KOA group (n = 10)				
Age (yr)		70.3	6.5	53-77
Sex (male/female)	1/9			
WOMAC pain score (out of 20)		4.6	2.2	2-8
	I	II	III	IV
K-L grade	2	3	2	3

system [22]–[24]. The structure of muscle synergy could reveal the relationship between musculoskeletal dynamics [25] and other biomechanical properties of the limb [26]. Based on these findings, muscle synergy analysis may be useful in musculoskeletal diseases that affect the biomechanics of the limbs.

However, few studies applied the muscle synergy analysis to musculoskeletal disease [27]. KOA and older adults may reduce muscle synergy due to simultaneous muscle co-contraction. To apply muscle synergy analysis in KOA, we formulated two aims: (1) to clarify whether there was a lower number of muscle synergies for both elderly individuals and those with KOA comared with young participants during gait; and (2) to clarify characteristic muscle synergies of KOA through a detailed analysis of muscle synergy components. Characterizing the changes in multiple muscle control associated with musculoskeletal degeneration may serve as a fundamental biomarker in joint disease.

II. MATERIALS AND METHODS

A. Subjects

Eleven young control subjects (YC), ten elderly control subjects (EC), and ten patients with KOA (KOA) volunteered as participants in this study. Patients with KOA were recruited from a local orthopedic clinic and were diagnosed by an orthopedic doctor. The severity of KOA was graded using the Kellgren and Lawrence grade (K-L grade) [28] based on radiographs. Both bilateral and unilateral KOA were included in this study. In the case of bilateral KOA, the more severe side in the K-L grade was defined as the measurement side. The participant demographics are shown in Table I. The presence of pain may alter the muscle activity pattern [29]. In order to avoid any effects of pain, an assessment of pain was conducted using the Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) [30]. The WOMAC consists of the following five factors: 1. Walking, 2. Stair Climbing, 3. Nocturnal, 4. Rest, and 5. Weight bearing. In this study, a pain score of 4.6 meant that subjects reported pain in any of the four factors, excluding the Walking factor. Therefore, subjects complaining of pain during gait were excluded from our study. The EC group completed the WOMAC as well. All subjects in the EC group had no pain in any factor. Moreover, individuals with a history of stroke, lower limb fracture, or pain during gait were excluded from the study. We explained the experiments in detail and obtained written consent from all subjects. All procedures used in this study were in accordance with the Declaration of Helsinki and were approved by the ethics review committee of Saitama Prefectural University (28507).

B. Experimental Data

Gait assessments were performed on a split-belt treadmill (Bertec, Columbus, OH, USA). To determine gait speed, all subjects performed gait at speeds from 1 km/h to 5 km/h (1 km/h, 2 km/h, 3 km/h, 4 km/h, 5 km/h) However, 4 km/h was difficult to achieve for many subjects, especially in many KOA cases. A recent study showed that the number of modules is affected by the gait speed [14]. In order to eliminate changes that are speed-dependent variations between subjects, it is necessary to have consistency of gait speed. In a previous study, it was shown that the gait speed of 2 km/h was the highest gait speed suitable for all patients with osteoarthritis without pain [31]. In our study, we chose a speed of 3 km/h, which could be performed by all subjects without pain. Before the experiment, the subjects were familiarized with moving on the treadmill at 3 km/h."

C. Data Collection

Surface electromyography (EMG) data from 16 muscles in the trunk and ipsilateral leg were collected at 1000 Hz by a wireless EMG system (Delsys Trigno Wireless System; DELSYS, Boston, MA, USA) with an inter-electrode distance of 10 mm. The measured leg was defined as the left leg for YC and EC, and the measured side for KOA. The 16 muscles included the paravertebral muscle (PVM), opposite paravertebral muscle (opp PVM), gluteus maximus (GMAX), gluteus medius (GMED), tensor fascia latae (TFL), adductor magnus (ADD), rectus femoris (RF), vastus medialis (VM), vastus lateralis (VL), semitendinosus (ST), biceps femoris (BF), peroneus longus (PL), tibialis anterior (TA), gastrocnemius medialis (MGAS), gastrocnemius lateralis (LGAS), and soleus (SOL).

A camera motion capture system (Vicon Motion Systems, Oxford, UK) was used to determine the spatial location of the body segments, as well as to calculate each joint angle during gait. The passive retroreflective markers were placed over bony landmarks according to the plug-in-gait full body functional model implemented in the camera motion capture system (bilaterally on the 2nd MTP head, heel, ankle, knee, thigh, anterior of shank and thigh, anterior superior iliac crest,

shoulder, upper arm, elbow, radial and ulna wrist, 2nd finger, forehead, and posterior head; single markers were placed on the jugular notch, inferior sternum, C7, T10, and right scapula). The data were collected at a rate of 100 Hz.

All data were synchronized using Vicon Workstation v4.5 software and saved to a disk for offline analysis.

D. Data Processing

Muscle activation signals were band-pass filtered (20–450 Hz) with a zero-lag fourth-order Butterworth filter, demeaned, rectified, and then smoothed with a zero-lag fourth-order low-pass (4 Hz) Butterworth filter [20]. In a previous study, it was reported that a slower cycle movement at 1 Hz should be accompanied by the use of a 5 Hz low-pass filter [32]. The gait speed of 3 km/h in this study is roughly equivalent to movement at 1 Hz. Therefore, the low-pass Butterworth filter at 4 Hz was used for smoothing based on a previous study [20]. Kinematic and kinetic data were low-pass filtered (6 Hz) with a zero-lag fourth-order Butterworth filter.

Each type of data was segmented into cycle-by-cycle sequences, with heel strike as the starting point. A gait cycle was defined as the interval between two consecutive heel strikes on the measurement side. For intra-subject comparisons, the data were time-interpolated over each gait cycle to fit a normalized 100-time point base. In addition, muscle activities from EMG data were normalized with reference to the peak values for each muscle activity. Each data profile of 10 continuous cycles was averaged to obtain the final profile of a gait cycle [32].

E. Extraction of Muscle Synergies

A non-negative matrix factorization (NNMF) was applied to the EMG data matrix to extract muscle synergies [12], [33], [34]. In this study, we defined a muscle module as each muscle synergy component. The EMG data matrices (M) were composed of 16 (number of muscles) × 100 (number of time points). These were decomposed into two components: spatial component (W) and temporal component (C). NNMF can be expressed as follows:

$$M \cong WC + e \tag{1}$$

where M (m × t matrix; m is the number of muscles and t is the number of time point of gait cycle) is a linear combination of spatial component W (m × n matrix; n is the number of module) and temporal component C (n × t matrix), and e is the residual error matrix. The number of muscle modules was determined using the variance account for (VAF). The VAF describes the amount of variability in the original EMG data accounted for by the EMG reconstructed from the temporal component and their spatial component [35]. To define the optimal module number, two criteria were used in this study. First, VAF was ≥90% for each of the 16 muscles [36]. Second, the inclusion of an additional module did not increase VAF by >5% [14], [20].

F. Comparison of Knee Joint Angles Between Groups

Knee angles were compared between the three groups to examine factors affecting changes in the number of modules. The knee joint angle was used in this study as it is clear that it is affected by KOA. In the present study, peak knee flexion angle in the stance phase was adopted.

G. Defined Muscle Module Groups

Sorting of muscle modules across each subject was first erformed on the basis of the peak timing of the temporal component of each muscle module, according to previous studies [36], [37].

H. Similarity in Muscle Modules

We performed NNMF analysis assuming four modules for all of the participants to assess how fewer modules would affect the result [20]. The definition of NNMF stipulates that intragroup modules must not be similar. In other words, the four modules used in the present study must all be independent of each other. To assess the independence between modules, cosine similarity was used ($r > 0.50$) [38]. Cosine similarity is a method of measuring the angle between two modules [34]. The lower the similarity, the closer the value is to zero. This means that each module is independent of the other. Conversely, the higher the similarity, the closer the value is to one. A high value of similarity means that each module is similar to the other. In other words, these modules may be extracted as a result of merging between the modules. In our study, values higher than 0.5 were defined as high similarity and values lower than 0.5 as low similarity. First, the similarities between the sorted modules within each group were compared to confirm if similar modules were extracted for all subjects within each group. Next, intergroup similarities of the sorted modules were compared. Finally, to determine whether each module within the group was independent, the similarities between the modules were compared.

Moreover, to determine muscle co-contraction, the number of active muscles per module was calculated. Active muscles in a module were defined as muscles that showed values exceeding 0.3 for the spatial component [21], [39]. This method has been used as an indicator of the number of co-contraction muscles contributing per module. For each module, the number of active muscles varied from 16 (i.e., all recorded co-contraction muscles) to 1 (i.e., no co-contraction muscles) [39]. The number of active muscles per module was calculated for the average spatial component in each group.

I. Statistics

Differences in the number of muscle modules among the three groups (YC, EC, and KOA) were evaluated by the Kruskal-Wallis test. Additionally, comparison of the knee flexion angles between the three groups was performed using the one-way analysis of variance (ANOVA) test. If a significant difference was found in the Kruskal-Wallis test and one-way ANOVA test, post-hoc multiple comparisons were performed to determine the relationships among the groups. Statistical significance was indicated by $p < 0.05$.

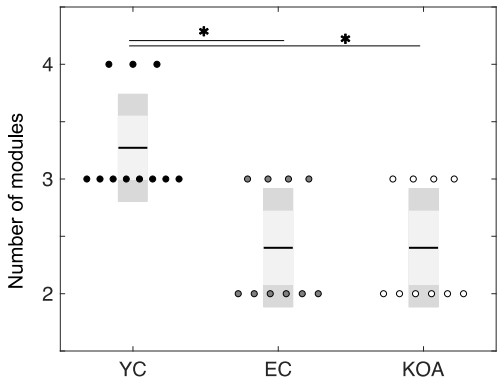

Fig. 1. Number of modules in each group (left: YC group, middle: EC group, right: KOA). Each dot shows the value for each subject. Each line in the box indicates the mean value for each data group. The light gray area and dark gray area in the box shows the standard deviation and 95% standard error of mean, respectively.

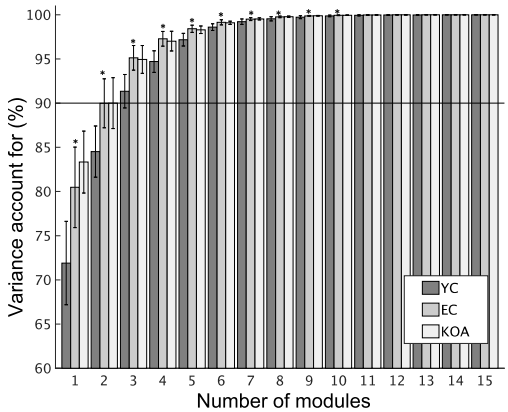

Fig. 2. The number of muscle modules selected accounted for >90% and additional module accounted for <5% of the VAF as depicted by the plot from all subjects (left: YC group, middle: EC group, right: KOA).

III. Results

A. Number of Muscle Modules

In the NNMF findings, three to four modules were extracted in YC group, whereas two to three modules were extracted in EC group and KOA group (Fig.1). The number of muscle modules was significantly different among groups. In particular, it was significantly lower in the EC and KOA groups than in the YC group ($p < 0.05$). The VAF was significantly lower in YC group compared to the EC and KOA groups from one to 10 modules (Fig.2).

B. Profiles for Each Muscle Module

Fig. 3 shows the temporal and spatial component in each group based on the results of the number of modules. The two to four modules were extracted in all subjects. The four modules were extracted in gait, which is consistent with previous studies [14], [40], [41]. Neptune *et al.* [41] have defined module contribution to the gait sub-task of body support as consisting of the following: forward propulsion and leg swing in the beginning of the early stance (∼15% gait cycle), the late stance (∼45% gait cycle), the early swing (∼70% gait cycle), and the late swing (∼85% gait cycle). In the present study, the four modules in the YC group were defined as the gait phases of early stance, late stance, early swing, and late swing, as they were consistent with these gait cycle [41]. Three modules in the YC, EC and KOA groups were extracted: two modules during the stance phase (early and late) and one module during the swing phase. Two modules in the EC and KOA groups were further extracted as one module during the stance phase.

To investigate how the merging of the modules would affect the results, four modules were extracted for all of the subjects in each group. Fig. 4 shows the temporal and spatial component in each module. A was categorized as YC-A, EC-A and KOA-A in the early stance phase, and B in the late stance phase. C was categorized in the early swing phase. Furthermore, D was categorized in the late swing phase. The four modules (A, B, C, and D) were defined as the gait phases, each of which showed results consistent with those of a previous study [41]. Furthermore, each module showed a high similarity between subjects within the group. one module during swing phase.

C. Comparison of Spatial Components of Each Module Group

The similarity between muscle modules was assessed using cosine similarity (0.50). Each value in Fig. 4 indicates the mean value of the similarity of each sorted module within groups. High values were showed within groups (YC: 0.81–0.66, EC: 0.91–0.62, KOA: 0.84–0.58). From these results, similar modules were extracted for all participants within each group by assuming four modules. Next, when comparing each module between groups, spatial components that assumed similar modules indicated high similarity (Table II; left). From this result, similar modules were extracted between groups (Module A: 0.74–0.62, Module B: 0.83–0.78, Module C: 0.69–0.63, Module D: 0.65–0.59).

On the other hand, the independence between modules within each group was indicated by the low similarity (Table II; right). The YC group showed independence in each module (A, B, C, and D). However, modules A–D and modules C–D showed higher levels of similarity. The EC and KOA groups showed less independence between modules A–D and C–D. Moreover, the KOA group was less independent in the modules A–B (YC: $r = 0.49$; EC: $r = 0.45$; KOA: $r = 0.57$) and B–C (YC: $r = 0.42$; EC: $r = 0.41$; KOA: $r = 0.51$), more so than in the YC and EC groups.

D. Number of Muscles in Each Module

The number of active muscles per module was calculated for each module group by determining the differences in spatial components (Table III). The cross mark shows the muscles exceeding 0.3 in the spatial component. In the YC group, the first module (YC-A) mainly represented the activation of hip muscle (GMAX, GMED and TFL) and knee extensors (RF, VM, VL). The second module (YC-B) was included in the activation of plantar flexor muscles (MGAS, LGAS and SOL). The third module (YC-C) represented some muscle contribution to swing of limb (ADD, TA and PVM). The fourth module (YC-D) mainly described a large contribution of the

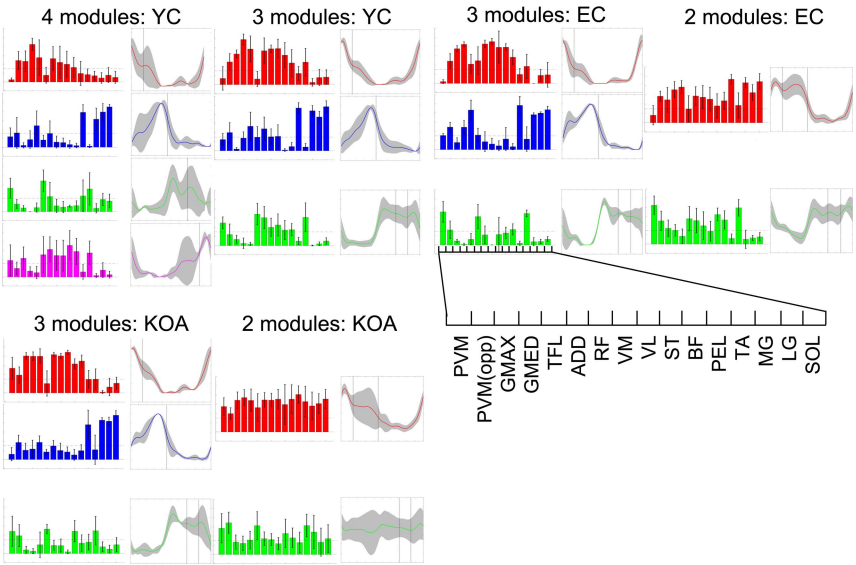

Fig. 3. The temporal and spatial component in each group based on the results of the number of modules. Each solid line (right; plot) and bar (left; bar plot) represents the value averaged across all subjects in each group. The gray scale (right; plot) and error bar (left; bar plot) show the standard deviation. Refer to Fig.1. for the number of subjects.

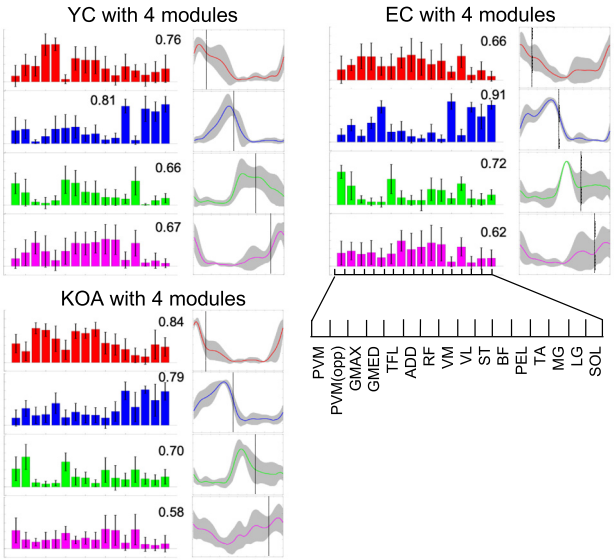

Fig. 4. The temporal and spatial component in each group when assuming four modules all of the subjects. Each solid line (right; plot) and bar (left; bar plot) represents the value averaged across all subjects in each group. The gray scale (right; plot) and error bar (left; bar plot) show the standard deviation. Each value shows the cosine similarity in each module.

hamstring muscles (ST, BF), hip flexior, and knee extensors (RF, VM, VL). Several different characteristics were observed in the EC and KOA groups compared to the YC group. In the EC and KOA groups, the first module (EC-A and KOA-A) were included in the activation of hamstring muscles (ST and BF). Furthermore, only KOA-A included the activation of plantar flexor muscles (LGAS and SOL).

E. The Peak Knee Flexion Angles Between Groups

Fig. 5 presents the comparison of peak knee flexion angles in the stance phase between groups. The EC and KOA groups displayed significantly greater knee flexion angles than the YC group ($p < 0.05$).

IV. Discussion

A. Summary

The aims of this study were as follows: (1) to clarify whether there was a lower number of muscle synergies for both elderly individuals and those with KOA compared with young participants during gait; (2) to clarify characteristic muscle synergies of KOA through a detailed analysis of muscle synergy components. The results showed significantly fewer modules in the EC and KOA groups than in the YC group. Next, in order to clarify what having fewer modules indicates, the similarity and the number of active muscles per module was investigated between modules. The results showed the merging of modules A-D, which may reflect age- or disease-specific muscle co-contraction for knee joint stability. However, the merging in modules A and B was shown only in the KOA group. It was indicated that the activity of the triceps surae muscle during the early stance phase is one of the characteristic patterns of muscle activity in KOA. Our results suggest that muscle synergy analysis may be useful in musculoskeletal diseases by focusing on the characteristic module structure within the spatial components, rather than comparing the number of modules.

B. Why Muscle Modules Merge: Modules A-D

Our results showed that the EC and KOA groups have fewer modules than the YC group (Fig.1). The fewer modules reflects the merging of module A-D (Table II; right), because the EC and KOA groups included the hamstrings in module A (Table III). These results showed that the fewer modules might reflect co-contraction between antagonistic muscles to support the lower limb during the early stance phase in the EC and KOA groups.

TABLE II
COMPARISON OF FOUR MODULE RESULTS IN EACH MODULE GROUP

Comparison of each module between groups			Comparison of between modules within each group			
A	YC-A	EC-A	YC group	A	B	C
EC-A	0.62		B	0.49		
KOA-A	0.74	0.71	C	0.43	0.42	
B	YC-B	EC-B	D	**0.55**	0.35	**0.55**
EC-B	0.81		EC group	A	B	C
KOA-B	0.78	0.83	B	0.45		
C	YC-C	EC-C	C	**0.53**	0.41	
EC-C	0.65		D	**0.64**	0.41	**0.57**
KOA-C	0.63	0.69	KOA group	A	B	C
D	YC-D	EC-D	B	**0.57**		
EC-D	0.65		C	**0.50**	**0.51**	
KOA-D	0.59	0.60	D	**0.57**	0.46	**0.59**

The threshold value was set to 0.5. The values higher than 0.5 were defined as high similarity and values lower than 0.5 as low similarity. In comparisons of between modules within each group (right), values in bold indicated a value higher than 0.5.

TABLE III
DETERMINING THE DIFFERENCES IN SPATIAL COMPONENTS

phase	Stance											Swing				
YC_A	×	×	×	×	×	×	×				×					
YC_B			×	×			×	×	×	×	×	×				
YC_C				×								×	×	×		
YC_D	×	×		×	×	×	×					×		×	×	×
EC_A	×	×	×	×	×	×	×					×		×	×	×
EC_B		×	×				×	×	×	×	×					
EC_C							×					×	×	×	×	×
EC_D		×		×	×	×	×						×	×	×	×
KOA_A	×	×	×	×	×	×			×	×	×		×	×	×	×
KOA_B			×	×			×	×	×	×	×			×		×
KOA_C							×					×	×	×	×	
KOA_D												×	×	×	×	×
Muscle	GMAX	GMED	TFL	RF	VM	VL	PVM(opp)	MGAS	LGAS	SOL	PEL	ADD	PVM	TA	ST	BF

The cross mark show the muscles exceeding 0.3 in the spatial components. A was categorized as module of YC-A, EC-A and KOA-A in the early stance phase, and B in the late stance phase. C was categorized in the early swing phase. Furthermore, D was categorized in the late swing phase. Each muscle was sorted in the order of activity from early stance to late swing.

In a previous study, KOA is characterized by knee joint instability with the deformation of the knee joint structure [42]. By assisting the non-contractile tissue (i.e., bone structure, ligament, meniscus) in maintaining and enhancing control, stiffness, and stability while performing various tasks, the musculature plays a significant role through co-contraction [43], [44]. Many previous studies have shown that KOA increased muscle activity and suggested the possibility of increased co-contraction of antagonistic muscle pairs in key lower extremity muscles during gait [5], [45], [46]. The lower muscle synergy in this study might be partially explained by compensatory movement with alterations of knee structure. On the other hand, the EC group also showed fewer modules and merging between module A-D. Many previous studies have indicated age-related changes in the co-contraction during walking, even among active, healthy elderly individuals [6]–[8], [47]. Many elderly individuals have associated sarcopenia, which is characterized by age-related loss of skeletal muscle mass and function [48]. Moreover, with aging, the complement of motor units undergoes a process of reduction and adaptation, which in turn affects the capacity to produce a force on the joints [9]. The maximal voluntary contractile (MVC) force in persons over 60 years of age is more than 25% lower than the MVC force in young

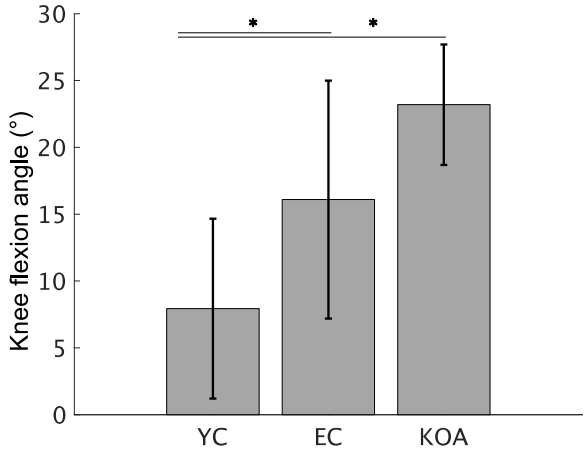

Fig. 5. Comparison to the peak knee flexion angle between groups (left: YC group, middle: EC group, right: KOA).The errorbar indicates the standar deviation.

adults in several muscles [49]. Thus, to exert the desired torque, the co-contraction of multiple muscles is necessary. For this reason, the EC group showed the fewer modules and merging between modules A and D.

Furthermore, we also found that module merging was associated with knee joint stability. A greater knee flexion angle at the stance phase was related to fewer modules (Fig. 5). This result suggests that a common kinematic feature of EC and KOA, reduced knee extension, is involved in merging between the modules A-D.

It is well known that the knee extension angle during gait is decreased in older adults compared to younger adults [50]. This may be due to an age-related decrease in the knee extensor muscle function associated with sarcopenia [48], such as a loss of quadriceps muscle volume [50] and a decline in quadriceps strength [1], [51]. Additionally, many studies have revealed that KOA is related to quadriceps muscle weakness [52]. For these reasons, aging and joint degeneration are considered to be common factors that hinder knee extension movement during gait.

The knee joint is more unstable in the flexion position than in the extension position. Dynamic stability of the knee joint is generated by an interplay between non-contractile tissues (i.e., bone structure, ligament, and meniscus) and contractile tissue (i.e., muscle). Regarding non-contractile tissues, knee joint stability is mainly obtained by the shape of bones, meniscus, and medial/lateral collateral ligament in the knee extension position. By contrast, knee flexion causes the loss of the stability with these non-contractile tissues without tension of the anterior/posterior cruciate ligaments. As a result, knee flexion induces an increase of muscle co-contraction. The reduction of modules may reflect a compensatory activity pattern in both KOA and EC that improves the stability in the knee flexion position.

Thus, we hypothesize that the fewer modules in EC and KOA was due to the merging between module A-D and muscle co-contraction. Furthermore, our study showed that this muscle co-contraction might cause the knee joint flexion position, a feature common to both EC and KOA.

C. The Difference Between EC and KOA

In our study, the EC and KOA groups showed similar results. However, some of the results showed characteristic alterations of the module structure in the KOA. It was the merging of module A and B. This feature is a modular structure specific to knee OA and may be a biomarker for detecting knee OA.

In our results, the KOA group showed lower independence in the moduel A-B (YC: $r = 0.49$; EC: $r = 0.45$; KOA: $r = 0.57$) more than the YC and EC groups (Table II; right). Moreover, module A included LGAS and SOL (Table III). The activity of the triceps surae muscle during the early stance phase is one of the characteristic patterns of muscle activity in knee OA. Heiden *et al.* had reported that the greater levels of muscle activation seen in KOA were associated with greater levels of lateral muscle activation (e.g., VL and LGAS) in early stance compared to controls of the same age [53]. The merging of module A-B in the results of this study may be an effect of the inclusion of LGAS and SOL in module A. These results were an effect that reflected the characteristic muscle activity pattern of KOA.

However, module A did not include the MGAS in our study. A previous study reported prolonged activity of the medial gastrocnemius compared to the anterior tibialis muscle in KOA patients [5]. In addition, it has been reported that in severe KOA the MGAS muscle is active for most of the gait cycle [3]. Contrarily, moderate KOA and asymptomatic individuals showed decreased MGAS activity during gait [46]. The reduction in MGAS activity may be an attempt to reduce joint loading, in particular, to reduce medial joint loading selectively in moderate and asymptomatic KOA. In the present study, the severity of KOA was not standardized (Table 1). Therefore, MGAS activity may not have consistently shown a typical pattern of muscle activity.

D. Muscle Synergy in Aging

A previous study reported that muscle synergies during walking were not affected by aging [40], in contrast to the findings of this study. This difference may be due to the different measurement protocols. While the previous study ensured a consistent cadence using a metronome, our study defined gait speed with a treadmill. Because the subjects in the previous study were restricted to cadence in addition to walking speed, the muscle synergy between the young and elderly participants did not change. Moreover, the previous study used principal component analysis (PCA) as a method to extract muscle synergy rather than NMF. NMF can present better results than rigorously orthogonal methods like PCA because muscle activation or neural spike shows a positive value [12], [33]. These factors possibly explain the differences between our findings and those of the previous study.

E. The Modules Dependent on Gait Speed

In the present study, many subjects across all three groups had three modules extracted (Fig. 1). In addition, the decrease in the number of modules to three modules represented a module deficit during the swing phase (Fig. 3). Moreover, because

of the slightly high similarity between the C-D modules in all of groups, the decrease in modules during the swing phase was an indication of merging between the C-D modules (Table II; right). This result differs from that of a previous study that clarified that gait is achieved from four to five modules [13]. This difference may be influenced by gait speed.

Previously, each module of muscle synergy had been thought to be consistent across gait speeds [13], [36], because primitive motor characteristics such as muscle synergy are hard-wired into the motoneuronal network involved in gait control [15]. However, a recent study showed that the number of modules changed depending on the gait speed [14]. In particular, a decrease in gait speed has been shown to reduce modules during the swing phase. Our selected gait speed (3 km/h) in the present study was consistent with the range of speeds that have been shown to reduce modules in the aforementioned study [14]. Therefore, the reduction to three modules and the merging modules during the swing phase may be a result of the influence of gait speed in all three groups.

In addition, the merging of modules A–D was indicated in the YC group ($r = 0.55$) (Table II; right), because the YC group included the RF, VM, and VL in module D (Table III). During a slow gait speed, the pendulum-like properties of the swinging leg can no longer be utilized for forward progression throughout the full swing phase [54]. Therefore, the RF, VM, and VL activity was shown during the swing phase and may have led to the merging of modules A–D.

F. Suggestion the Usefulness of Muscle Synergy Analysis for Musculoskeletal Disorder

Our results suggest that muscle synergy analysis may be useful in musculoskeletal diseases by focusing on the characteristic module structure within the spatial components, rather than comparing the number of modules. Most recent studies attempted to apply muscle synergy analysis to neurological disorders. Clark *et al.* reported that patients with post-stroke hemiparesis have reduced modules during gait compared to controls [20]. Moreover, in patients with Parkinson's disease, fewer modules were required during gait compared to healthy older adults. These reports of reduced modules in these previous studies reflect the co-contraction of muscles associated with abnormal muscle tone [20]. Therefore, it may be useful as an indicator of disease severity. However, it is difficult to identify underlying neurological disorders from the changes in module structure, because patients with neurological disorders show a wide variety of symptoms regardless of the location of the disorder. Contrarily, in musculoskeletal diseases, it may be possible to extract disease-specific module structure by using muscle synergy analysis. Musculoskeletal disorders cause any kind of abnormal movements with joint dysfunction. Co-contraction and prolonged activity of muscles are some of the most characteristic changes. Patients with anterior cruciate ligament injury walk with prolonged duration of hamstrings muscle activity [55]. These changes in muscle activity may affect muscle synergy analysis. Patients with anterior cruciate ligament injury showed different muscle synergy component compared to non-injured subjects [27]. Some specific muscle activities have also been shown in hip osteoarthritis. In a previous study, patients with hip osteoarthritis reported increased activity for gluteus medius, tensor fascia latae, and semitendinosus muscles [56]. And, another previous study reported altered muscle activity of the tibialis anterior muscle in hip osteoarthritis [57]. Thus, although muscle synergy may be altered in hip osteoarthritis, it has yet to be applied to hip osteoarthritis in research.

The results of this study have shown that a gait speed of 3 km/h could cause fewer modules during the swing phase. Many musculoskeletal diseases are associated with pain and joint dysfunction, which decrease gait speed. In a previous study, it was shown that the highest gait speed without pain and loss coordination in patients with osteoarthritis was 2 km/h [31]. Therefore, it is possible that there are fewer modules during the swing phase in patients with musculoskeletal diseases.

In this context, the fewer modules in the stance phase may be a disease-dependent change that can occur in any musculoskeletal disease with co-contraction and prolonged muscle activity. This is similar to patients with neurological disorders in which co-contraction of muscles results from abnormal muscle tone. Therefore, a comparison of the number of modules alone cannot reveal the anomalies underlying musculoskeletal disease. However, by quantifying the muscle activity contained within the modules, it may show a disease-specific module structure. In our study, we have been able to provide an indication of neuromuscular features in KOA patients. This finding demonstrated the potential application of muscle synergy analysis to other musculoskeletal diseases.

G. Limitations

The present study has some limitations. First, we could not clarify the differences in the number of modules between the EC and KOA groups. The difference between both groups may affect the activation level of muscle co-contraction. A previous study hypothesized that some smaller co-contraction levels tend to stabilize the system, and higher levels of co-contraction were not particularly efficacious [43]. Thus, the co-contraction in the EC group may be less compared to KOA. However, the co-contraction in the KOA group may have shown higher activity than that in the EC group. The difference in these co-contraction levels may reflect the characteristic muscle co-ordination found in KOA. However, in this study, which was normalized by the peak value in the gait cycle of each muscle, it was not possible to evaluate the activity level. Additional studies are needed to clarify the characteristic multiple muscle activation patterns in KOA, using normalization at maximum voluntary contraction. Second, the sex of the participants could not be controlled. All participants in the YC group were male, whereas the majority of the KOA group were female. In several studies of healthy young individuals, the subjects were all male [14]. In order to compare the EC and KOA groups in the present study, we considered that the control group (YC) needed to show results that were consistent with those of the previous study [14]. Therefore, only males were recruited for the YC group. On the other hand, the KOA group consisted of many female participants. A previous study reported that the prevalence of KOA was

higher in females than in males in Japan [58]. Therefore, the number of female participants was higher than the number of male subjects. Finally, this study could not determine the relationship between KL grade and the merging of modules due to the small number of subjects. Therefore, it is necessary to increase the number of subjects in future studies to determine the relationship between KL grade and the merging of modules, and to improve gender imbalance.

The cross mark show the muscles exceeding 0.3 in the spatial components. A was categorized as module of YC-A, EC-A and KOA-A in the early stance phase, and B in the late stance phase. C was categorized in the early swing phase. Furthermore, D was categorized in the late swing phase. Each muscle was sorted in the order of activity from early stance to late swing.

V. Conclusion

Our study found fewer muscle modules in both the elderly and KOA groups due to the increase of muscle co-contraction. By focusing on the module structure, our results showed that the merging of modules A and B is characteristic in KOA. This indicates that muscle synergy analysis may be useful in future studies exploring other musculoskeletal diseases characterized by focusing on the abnormalities of components in muscle activity. As such, describing the changes in multiple muscle control associated with musculoskeletal degeneration can serve as a fundamental biomarker in joint disease.

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